

Edward Lee

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EDUCATION

University of California, San Diego

Masters of Science in Engineering Sciences (Controls & Mechatronics)

GPA: 4.0 | Expected March 2027

- **Coursework:** Bioinspired Mobile Robots, Robot Motion Planning, Robotic Estimation, Soft Robotics

Bachelors of Science in Mechanical Engineering

GPA: 3.6 | Graduated June 2025

- **Coursework:** Mechanical Design, Product Design, Project Management, Statics, Dynamics, Solid Mechanics, Fluid Mechanics, Thermodynamics, MATLAB, Mechanics of Materials, Autonomous Vehicles, Linear Controls

EXPERIENCE

Matic Robots

December 2025 – January 2026

Robot Production Intern (9-Week Internship)

- Assembled 720+ electromechanical subsystems for autonomous robots, ensuring precise mechanical fit, alignment, and adherence to torque specifications by strictly following engineering documentation
- Executed production-level functional testing to identify tolerance and assembly failures, diagnosing and reworking 40+ units while providing Design for Manufacturing (DFM) feedback to engineering
- Reduced assembly cycle time by 10% by applying bottleneck analysis and Work-In-Process (WIP) management strategies to eliminate any idle time and to streamline the entire assembly process

Advanced Robotics and Controls Lab

September 2025 – Current

Graduate Research Assistant

- Designing and prototyping novel robotic mechanisms, including a 2-DOF planar parallel platform to replicate scapular motion and a spherical parallel manipulator to model shoulder joint kinematics, increasing fidelity in biomechanics testing
- Owned electromechanical integration of multi-axis robotic subsystems, designing motor mounts, shaft couplings, and packaging constraints while coordinating CAN-based BLDC drives and embedded controllers for reliable assembly and testing

Perrybotics VEX U

July 2023 – May 2024

Co-Founder & Mechanical Engineer

- Co-led the design, fabrication, and testing of a competitive 15" x 15" x 6" VEX U robot, coordinating subsystems including drivetrain, scoring, climbing, and autonomous operation under strict size, weight, and performance constraints for competition
- Identified and redesigned an inefficient catapult mechanism, replacing it with a custom deployable climbing hook and pulley system (FDM 3D printing and CNC machining), improving SWaP by 9% and enhancing scoring efficiency by 31%
- Programmed the robot's autonomous routine in C++, applying motion control and PID tuning to coordinate drivetrain and mechanism actuations, and verified performance through on-field sensor feedback and iterative testing for reliable scoring

PROJECTS

PIBBL: Personal Intelligent Bipedal Bot for Locomotion

September 2025 – Current

Robotics Engineer (Personal Project)

- Designed and built a wheeled bipedal robot to compare classical control techniques with simulation-based approaches while experimenting with wheel-leg hybrid locomotion for improved maneuverability and step-climbing capability
- Integrated a 9-DOF IMU and high-resolution motor encoders to enable real-time state estimation, closed-loop balance control, and reliable torque and velocity feedback for the servo actuation system
- Developed and iterated a modular Python control architecture using object-oriented programming and a multi-state control machine that transitions through initialization, idle, rise-to-stand, balancing, and disturbance-recovery modes

RoboticARM v1

December 2022 – April 2023

Robotics Engineer (Personal Project)

- Designed and fabricated a 5-DOF robotic arm and a 5-DOF robotic hand module, beginning with hand sketches and hand calculations to SolidWorks modeling and FDM 3D printing for rapid prototyping and iterative refinement of mechanisms
- Determined the need for a 4:1 gear ratio to deliver 7.2 Nm joint torque through analytical calculations, then verified performance with a von Mises stress analysis in SolidWorks Simulation and repetitive physical testing
- Implemented an open-loop motion control system with joint angle calibration to demonstrate real-world operation, remotely commanding the arm to execute pick-and-place tasks while validating mechanical reach and load handling

SKILLS

- **Technical Skills:** SolidWorks, Finite Element Analysis, GD&T, DFM/DFA, Wire Harness Design, Python, C++, OOP, State Machines, Robotics, ROS2, Embedded Systems, Kinematics, PID, Rapid Prototyping, 3D Printing, CNC Machining